

The Hill Cipher

Five words: “*The matrix is the key*”

The Hill cipher encrypts m plaintext elements $(x_1, x_2, \dots, x_m)$ into m ciphertext elements $(y_1, y_2, \dots, y_m)$ by multiplying by an m by m matrix-key K . Note that K must have an inverse, K^{-1} , for decryption to be possible. If we are using elements from $\mathbb{Z}_{26}$ then:

$$\begin{aligned}(y_1, y_2, \dots, y_m) &= (x_1, x_2, \dots, x_m)K \bmod 26 \\ (x_1, x_2, \dots, x_m) &= (y_1, y_2, \dots, y_m)K^{-1} \bmod 26\end{aligned}$$

Example:

Encrypt (9,20,11,18) using $K = \begin{bmatrix} 11 & 8 \\ 3 & 7 \end{bmatrix}$

K is a 2 by 2 matrix, so we encrypt $m=2$ elements at a time.

$$\begin{bmatrix} 9 & 20 \end{bmatrix} \begin{bmatrix} 11 & 8 \\ 3 & 7 \end{bmatrix} = \begin{bmatrix} 159 & 212 \end{bmatrix} \bmod 26 = \begin{bmatrix} 3 & 4 \end{bmatrix}$$

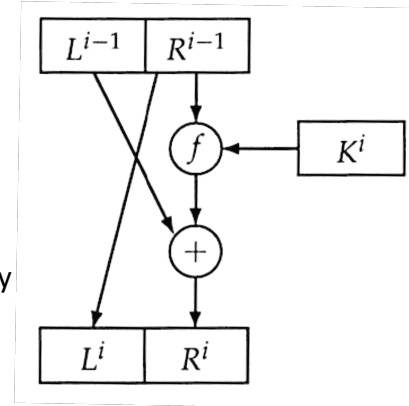
$$\begin{bmatrix} 11 & 18 \end{bmatrix} \begin{bmatrix} 11 & 8 \\ 3 & 7 \end{bmatrix} = \begin{bmatrix} 175 & 214 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 & 6 \end{bmatrix}$$

The cipher text is: (3,4,19,6)

The Feistel Cipher

One round of a Feistel cipher:

- Split the previous state into two sides L^{i-1} and R^{i-1} .
- Set $L^i = R^{i-1}$
- Set $R^i = L^{i-1} \oplus f(R^{i-1}, K^i)$ where K^i is the round key
- Return $(L^i R^i)$



DES is a 64-bit, 16-round, Feistel cipher with an “interesting” encryption function f .

In DES R^{i-1} would be 32-bits and the encryption function $f(R^{i-1}, K^i)$ would:

- Expand R^{i-1} from 32-bits to 48-bits.
- XOR the expanded R^{i-1} with the 48-bit round key K^i and write the result as eight 6-bit strings.
- Map each 6-bit string to 4-bits using 8 different S-boxes (yes... 8 of them in total!)
- Permute the resulting 32-bit string using a defined permutation P .

Let’s consider an 8-bit Feistel cipher with a 4-bit encryption function f , 4-bit round key, and a permutation P on 4-bits (and skip the expansion and the S-boxes!):

$$f(R^{i-1}, K^i) = P(R^{i-1} \oplus K^i)$$

Example:

The current state of an 8-bit Feistel cipher is 10110011. Compute the next state using $f(R^{i-1}, K^i) = P(R^{i-1} \oplus K^i)$, where $P = \{2, 4, 1, 3\}$, with round key 1001.

$$L^i = 0011$$

$$\begin{aligned}
 R^i &= 1011 \oplus f(0011, 1001) \\
 &= 1011 \oplus P(0011 \oplus 1001) \\
 &= 1011 \oplus P(1010) \\
 &= 1011 \oplus 0011 \\
 &= 1000
 \end{aligned}$$

The next state is 00111000

Exercises:

1. Decrypt $(3,4,19,6)$ using the Hill Cipher with $K^{-1} = \begin{bmatrix} 7 & 18 \\ 23 & 11 \end{bmatrix}$.

Show that $K \times K^{-1} \bmod 26 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ where $K = \begin{bmatrix} 11 & 8 \\ 3 & 7 \end{bmatrix}$.

2. The current state of an 8-bit Feistel cipher is 11110001. Compute the next state using $f(R^{i-1}, K^i) = P(R^{i-1} \oplus K^i)$, where $P = \{4,1,2,3\}$, with round key 0101.

Answers:

1. $[113\ 98] \bmod 26 = [9\ 20]$
 $[271\ 408] \bmod 26 = [11, 18]$
(9,20,11,18)

$$\begin{bmatrix} 261 & 286 \\ 182 & 131 \end{bmatrix} \bmod 26 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2. 00011101